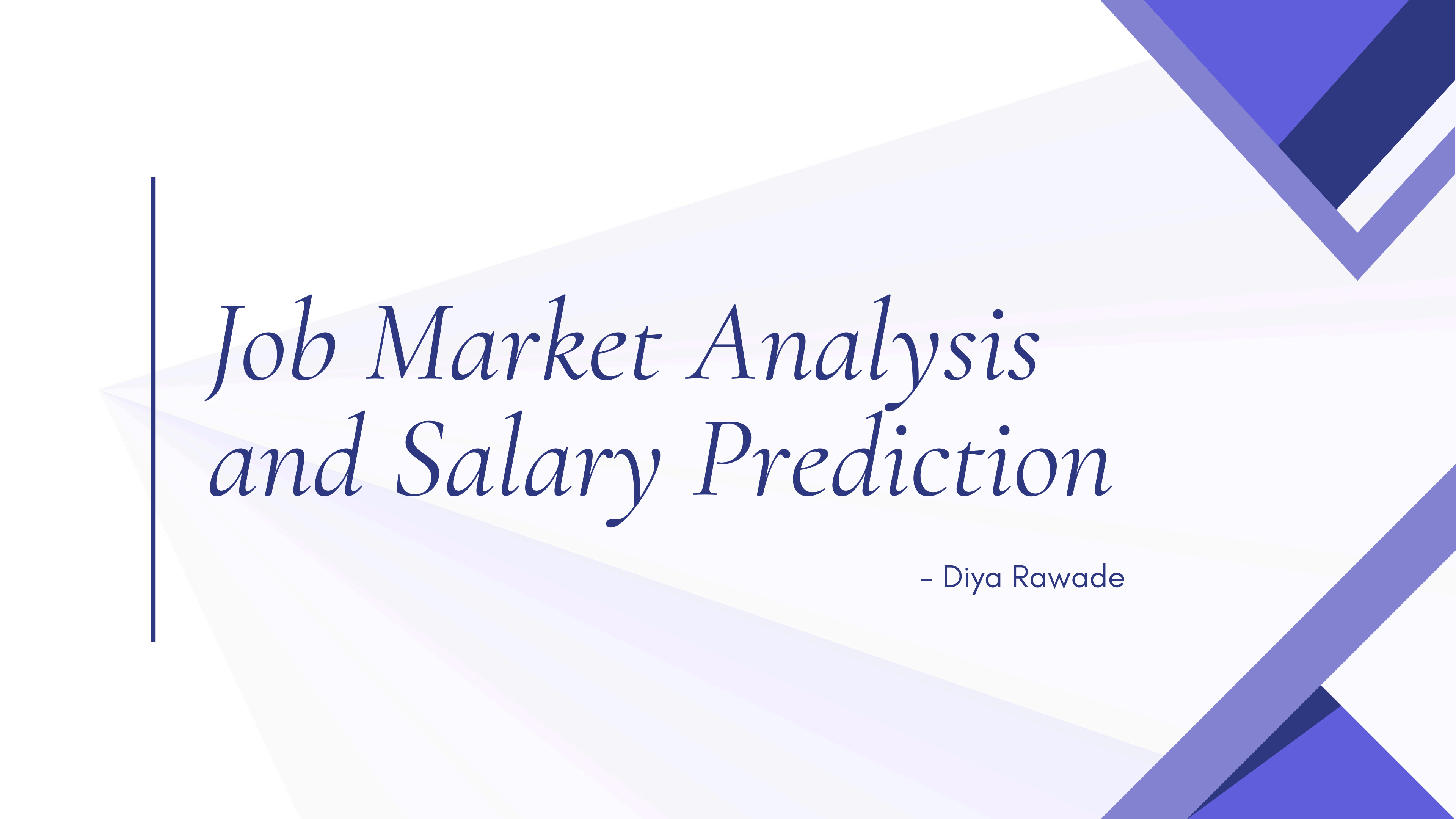




Job Market Analysis and Salary Prediction

– Diya Rawade

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Problem Statement



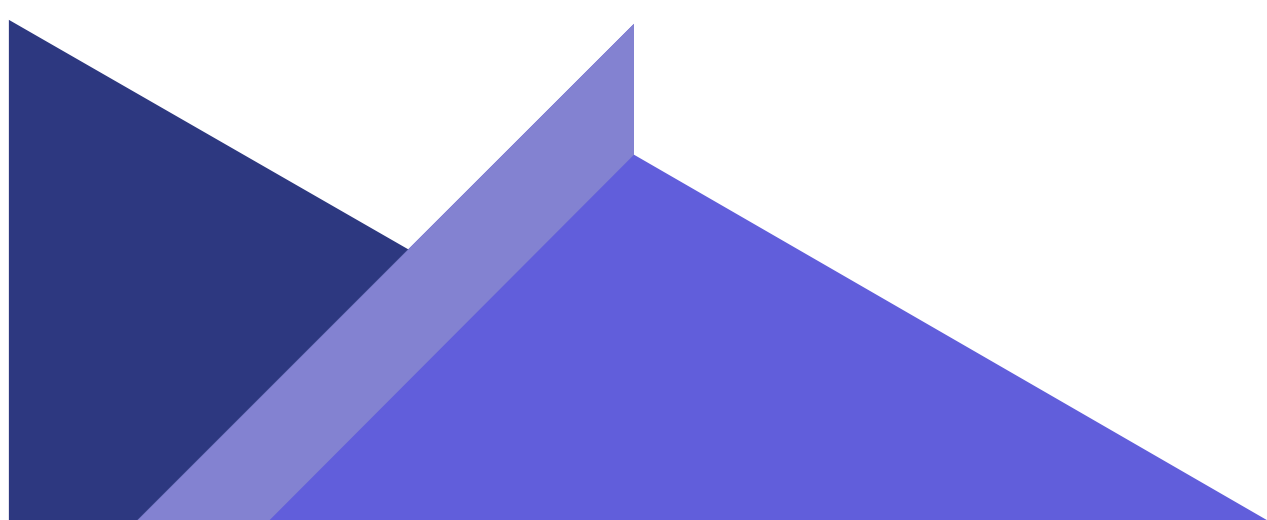
Salary estimation is complex and depends on many factors.



While manual estimation is often inaccurate a data-driven approach improves accuracy. This helps job seekers understand expected salaries.



Companies can use it for fair salary distribution as it increases transparency in the job market.



Dataset Description

- job_title, experience_level — Job title tells what role the person has, and experience level tells their seniority (junior/senior etc.).
- company_location, size — Location tells the country of the company; size tells whether it's a small, medium, or large company .Both affect salary.
- salary_in_usd — This is what the model tries to predict, the actual salary is converted to USD for consistency.
- Categorical and numerical data — Some columns have text values (like job title ,employee residence), and some have numbers (like salary) both types of data are present.
- Countries and roles — The dataset is diverse, covering many job roles and countries, making the model more generalizable.
- This dataset is the foundation. It is used for training ML model .The model learns salary patterns from it.



Data cleaning and EDA

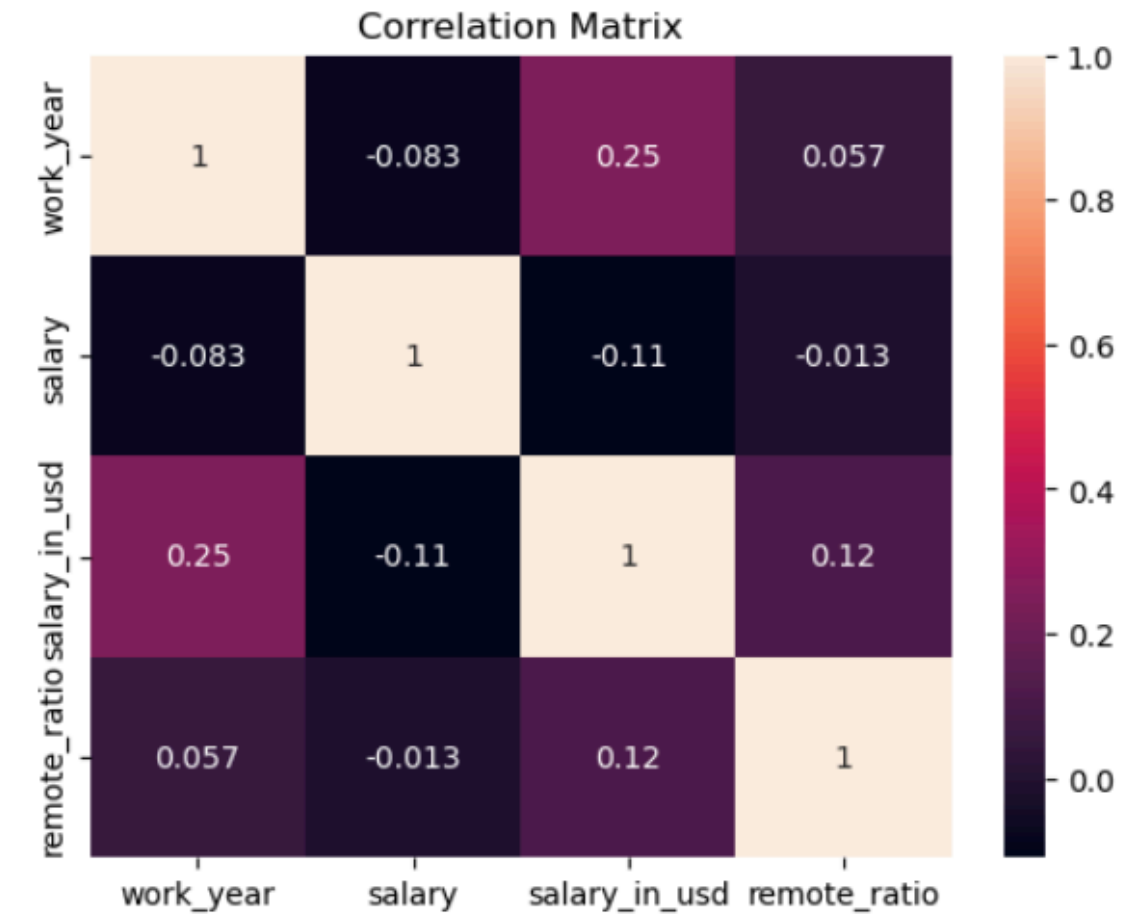
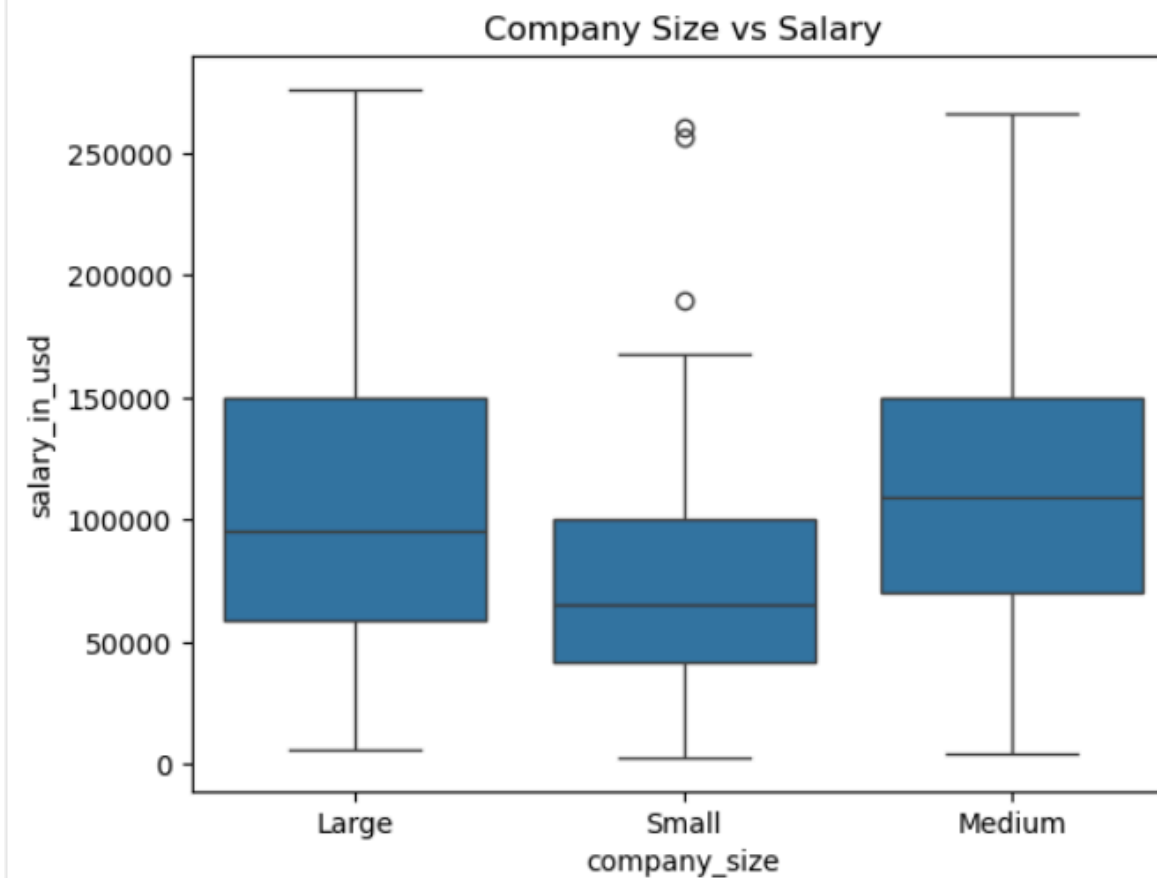
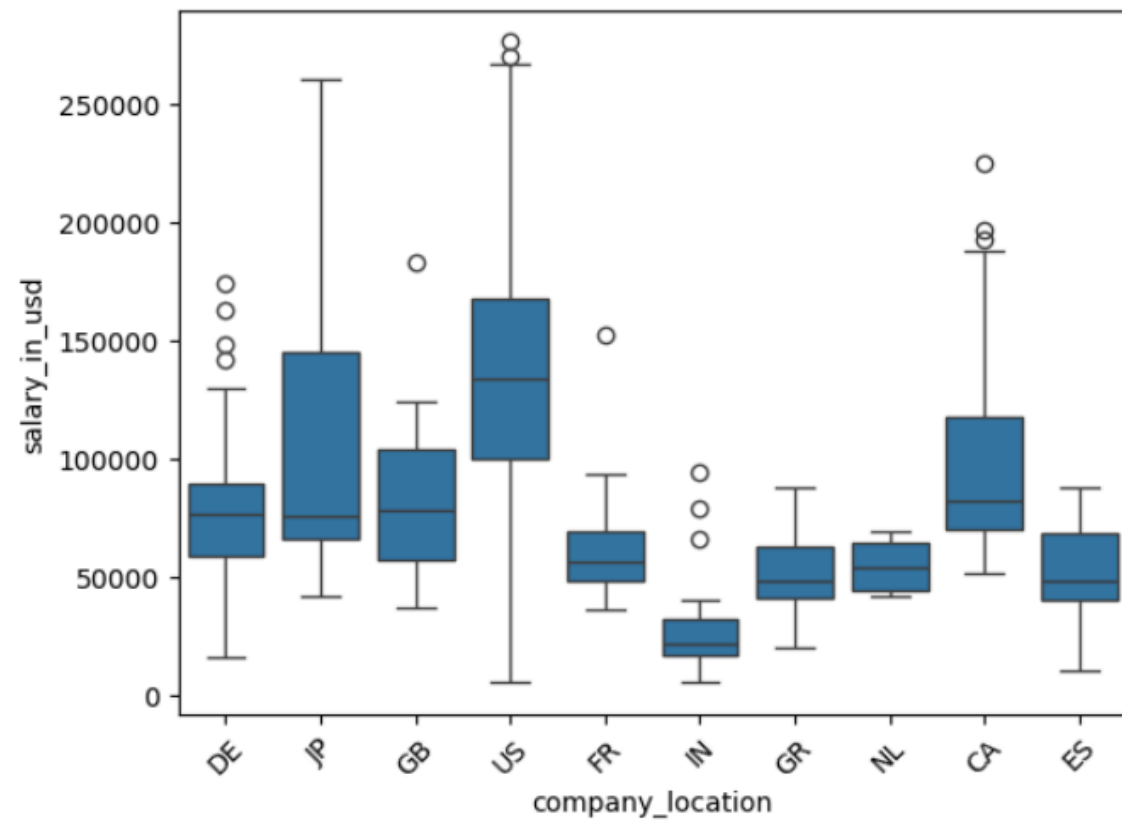
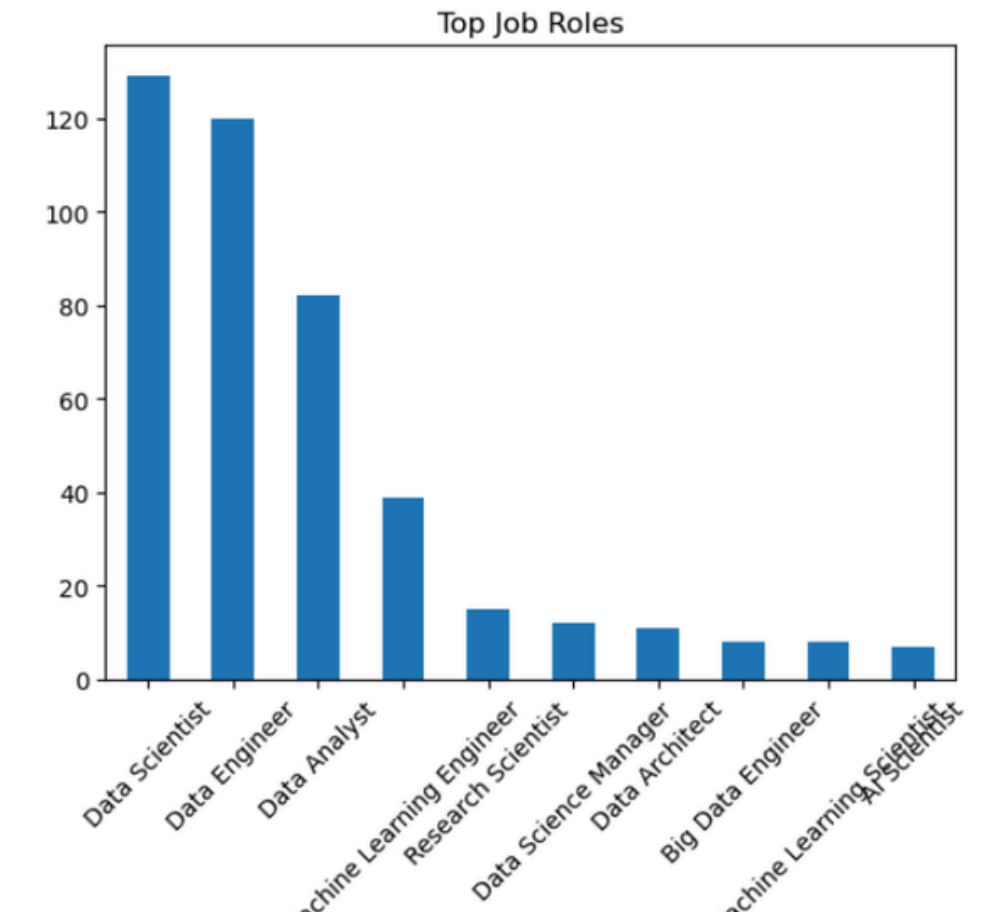
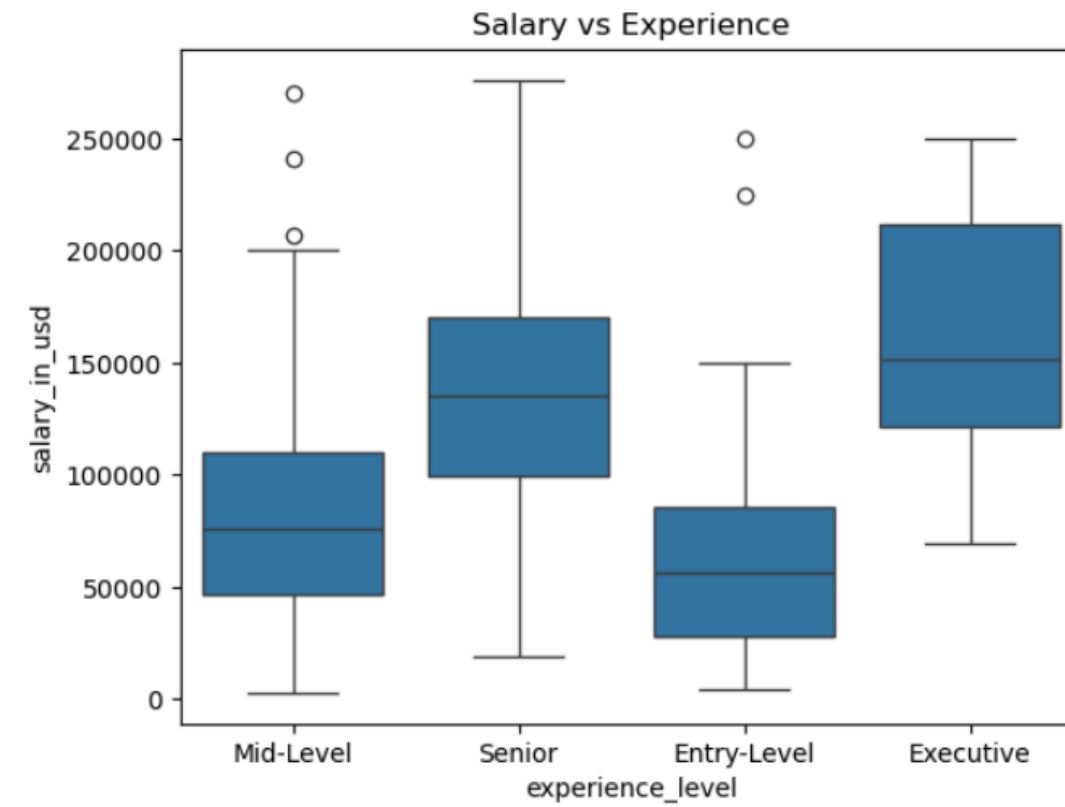
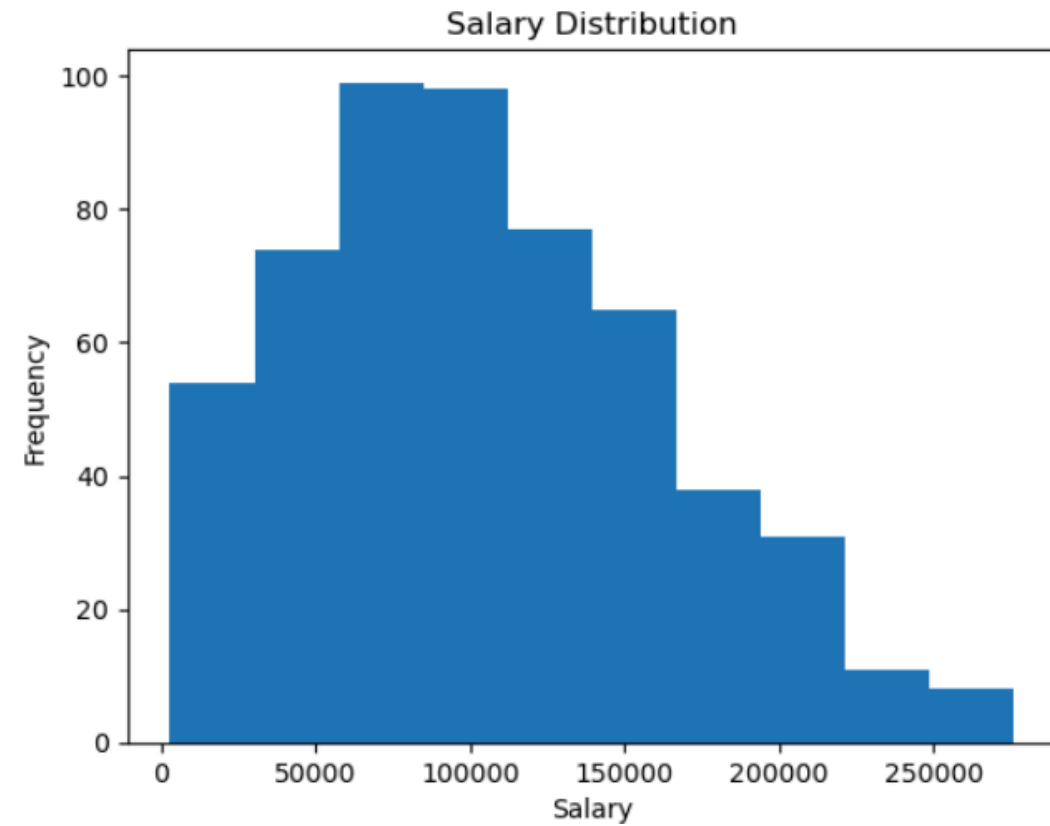
DATA CLEANING -

- We removed duplicate records and handled missing values to ensure data quality.
- Dropped unnecessary columns like salary_currency to simplify the data.
- Converted all salaries from different currencies to USD so that the model works on a uniform scale.
- Ensured consistent data types across columns.

EXPLORATORY DATA ANALYSIS -

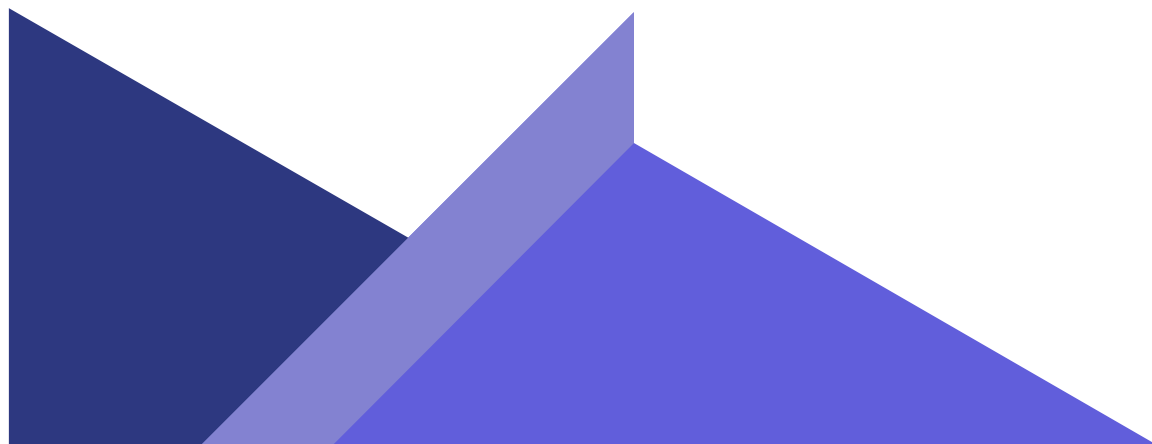
- Here, we analyzed salary distribution.
 - Studied salary vs experience- We compared salary across different experience levels to see if more experience leads to higher pay.
 - Checked job-role frequency and analyzed location based salaries- how salaries differ by country or city, since location greatly impacts pay.
We used visualizations for insights.
- 

Visualizations



Feature Engineering

- We converted categorical data into numerical form as text-based columns can't be directly fed into ML model.
- Used One-Hot Encoding (creates separate binary (0/1) columns for each category) instead of Label Encoding (assigns numbers like 1, 2, 3 to categories, which can wrongly imply an order e.g., "Manager > Analyst"). One-Hot avoids this.
- After encoding, all data is numeric and suitable for input into ML algorithms.
- Properly encoded features give the model better information, which leads to more accurate salary predictions.



Model Building and Evaluation

- First, we split dataset into training and testing data.
- Used 80% training, 20% testing — 80% of data is used to teach the model; the remaining 20% is used to evaluate its performance on unseen data.
- We trained Random Forest model and also Linear Regression.
- We used R^2 Score for accuracy — R^2 (R-Squared) measures how well the model explains the variation in salary. A value close to 1 means the model is very accurate.
- We used MAE for error measurement — MAE (Mean Absolute Error) tells us the average difference between predicted and actual salary. Lower MAE = better model.
- Random Forest gave better results — It had a higher R^2 score and lower MAE.
- Based on these evaluation metrics we chose Random Forest as final model.



Feature Importance

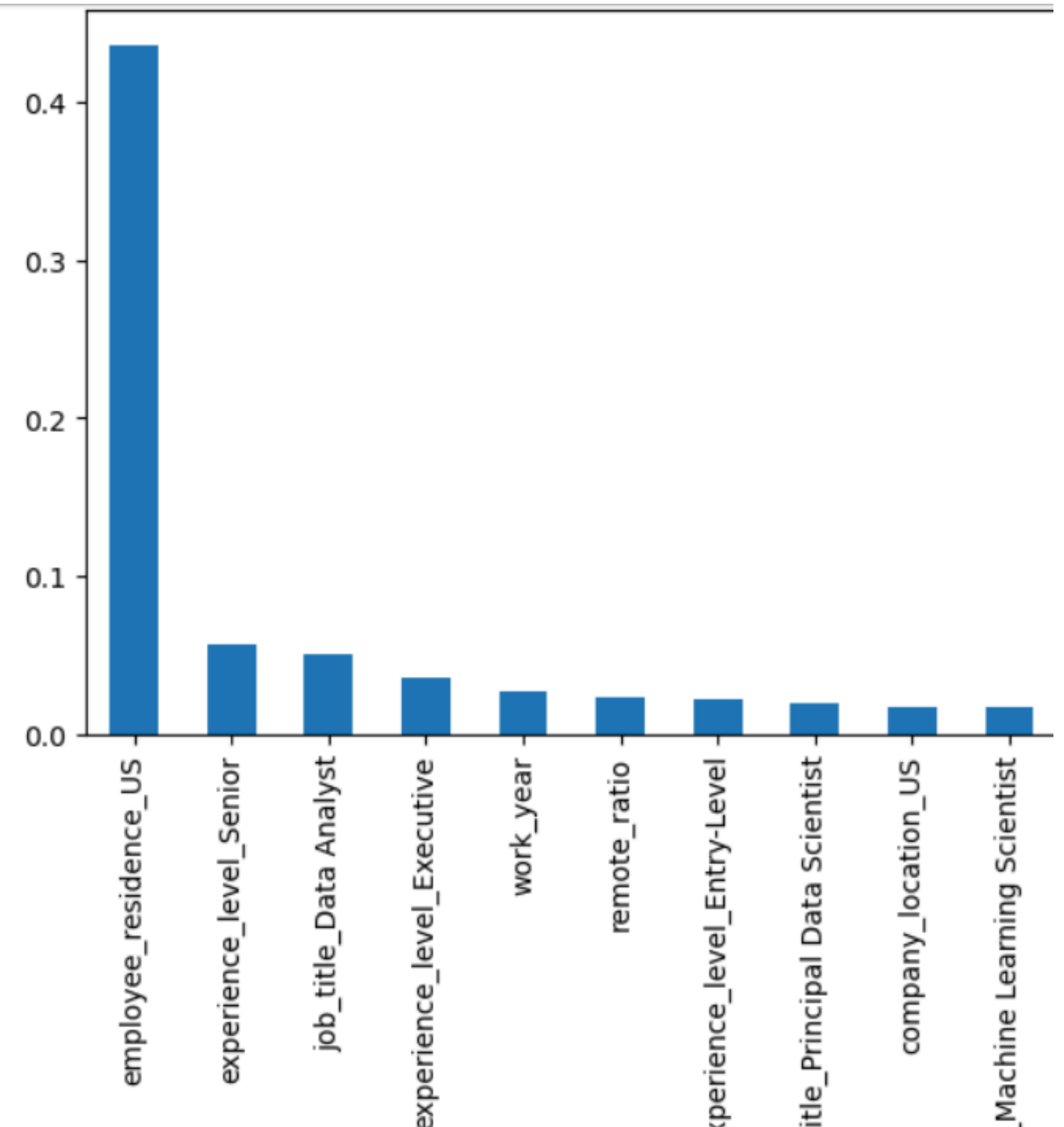
We used a bar chart for visualization which ranks all features by its importance score making it easy to understand.

From the feature importance graph, we can observe that employee residence and job title have the highest impact on salary prediction, followed by experience level. This indicates that location and job role are major factors affecting salary.

- employee_residence → highest (~0.48)
- job_title → second (~0.20)
- experience_level → third (~0.12)

Others → smaller impact

Since categorical variables were label encoded, the importance may not perfectly reflect real-world hierarchy.



Web Application

- We used joblib to save and load the trained model. It helps in model persistence, allowing us to reuse the trained model without retraining making it faster and efficient.
- Then we used streamlit to build an interactive web application. We created user input interface using dropdowns and sliders to ensure users select valid inputs reducing errors.
- Once the user clicks predict, the trained ML model processes the inputs and returns a salary estimate.



Job Salary Prediction App

Predict salaries base on job details

Work Year

2020

- +

Experience Level

Entry-Level

▼

Employment Type

Full-Time

▼

Job Title

Data Scientist

▼

Employee Residence

US

▼

Remote Ratio



Thank You